

A STEINER TRIPLE SYSTEM OF ORDER 13 WHOSE BURNING NUMBER INCREASES ON DOUBLING AND ON TRIPLING

ABSTRACT. Burgess, Danziger, Jones, Marbach and Pike ask, in the Discussion of [1], whether the projective and affine triple systems are the only non-degenerate Steiner triple systems whose burning number increases on doubling or tripling. We exhibit a counterexample. Let H be the cyclic STS(13) generated by the base blocks $\{0, 1, 4\}$ and $\{0, 2, 7\}$ modulo 13. Then H is non-degenerate and, its order being neither of the form $2^{n+1} - 1$ nor of the form 3^m , it is neither projective nor affine; yet $b(H) = 5$ while $b(D) = 6$ for its double-and-add-one image D , an STS(27), and $b(T) \geq 6$ for its tripling image T , an STS(39). The values at orders 13 and 27 are forced for *every* Steiner triple system of those orders by two unconditional bounds of [1] itself, so no computation is needed; they are also obtained directly from the definition of the burning process. The refutation is thus arithmetic: it turns on the orders alone, and it says nothing about orders at which those two bounds do not coincide.

1. THE QUESTION

A *Steiner triple system* of order v , written STS(v), is a pair $(V, \mathcal{B})$ with $|V| = v$ and $\mathcal{B}$ a set of 3-subsets of V such that every unordered pair of points lies in exactly one block; such a system exists precisely when $v \equiv 1$ or $3 \pmod{6}$, and then $|\mathcal{B}| = v(v-1)/6$.

Burning a Steiner triple system [2, 1] is the following one-player round-based process. Let $F(0) = \emptyset$. In round $r \geq 1$ the arsonist ignites one point $u_r \notin F(r-1)$, and *simultaneously* every point outside $F(r-1)$ that lies in a block whose other two points are both in $F(r-1)$ catches fire; writing ∂S for the set of such points,

$$(1) \quad F(r) = F(r-1) \cup \partial F(r-1) \cup \{u_r\}.$$

The propagation in (1) is a *single* step, not a closure. The *burning number* $b(H)$ is the least k for which some choice of sources $u_1, \dots, u_k$ gives $F(k) = V$. In the *lazy* variant the fire spreads to a fixpoint and no source is added after the first round, so the lazy burning number $b_L(H)$ is the least size of a set of points whose closure under “the third point of the block through a pair” is all of V . A subsystem of H is a subset of V closed under that operation.

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Following [1], H is *non-degenerate* if every maximal proper subsystem S of H satisfies $b_L(S) = b_L(H) - 1$.

Two classical recursions appear in the question. *Doubling* (double-and-add-one, [3, Construction 2.15]) takes an STS(v) on V , a set W of $v + 1$ further points and a 1-factorization $F_0, \dots, F_{v-1}$ of the complete graph on W , and produces the STS($2v + 1$) on $V \cup W$ whose blocks are those of the original together with $\{x\} \cup e$ for every $x \in V$ and every $e \in F_x$. *Tripling* ([3, Construction 2.17]) produces an STS($3v$) on $V \times \mathbb{Z}_3$ whose blocks are the v vertical triples $\{(x, 0), (x, 1), (x, 2)\}$ together with $\{(x, i), (y, j), (z, k)\}$ for every block $\{x, y, z\}$ of the original and every $(i, j, k) \in \mathbb{Z}_3^3$ with $i + j + k \equiv 0$. The projective triple systems PG($n, 2$), of order $2^{n+1} - 1$, are exactly those obtained by repeated doubling from STS(3); the affine triple systems AG($m, 3$), of order 3^m , are exactly those obtained by repeated tripling from STS(3).

The question refuted here stands in the Discussion of [1] and carries no number there, so we quote it from the e-print source together with the sentence before it, the two internal cross-references redacted:

Theorems [...] and [...] also show that doubling a projective triple system or tripling an affine triple system increases the burning number by one. We ask if these are the only non-degenerate systems whose burning numbers increase on doubling or tripling.

We read “increase on doubling *or* tripling” in its union sense, which is the weaker target.

Theorem 1. *Let H be the STS(13) on $\mathbb{Z}_{13}$ whose 26 blocks are listed in (2), let D be the STS(27) with block set (3), obtained from H by doubling, and let T be the STS(39) obtained from H by tripling. Then*

- (a) H is non-degenerate;
- (b) $13 \neq 2^{n+1} - 1$ for every $n \geq 0$ and $13 \neq 3^m$ for every $m \geq 0$, so H is neither a projective nor an affine triple system;
- (c) $b(H) = 5$, $b(D) = 6$ and $b(T) \geq 6$.

So H is a non-degenerate Steiner triple system, neither projective nor affine, whose burning number increases on doubling and increases on tripling. With “or” read as a union, the quoted question therefore has answer no.

2. THE WITNESS

Let H be the cyclic STS(13) on $\mathbb{Z}_{13}$ obtained by developing the base blocks $\{0, 1, 4\}$ and $\{0, 2, 7\}$ modulo 13. Its 26 blocks are

$$\begin{aligned}
 (2) \quad & \{0, 1, 4\}, \{0, 2, 7\}, \{0, 3, 12\}, \{0, 5, 11\}, \{0, 6, 8\}, \{0, 9, 10\}, \\
 & \{1, 2, 5\}, \{1, 3, 8\}, \{1, 6, 12\}, \{1, 7, 9\}, \{1, 10, 11\}, \\
 & \{2, 3, 6\}, \{2, 4, 9\}, \{2, 8, 10\}, \{2, 11, 12\}, \\
 & \{3, 4, 7\}, \{3, 5, 10\}, \{3, 9, 11\}, \\
 & \{4, 5, 8\}, \{4, 6, 11\}, \{4, 10, 12\}, \\
 & \{5, 6, 9\}, \{5, 7, 12\}, \\
 & \{6, 7, 10\}, \{7, 8, 11\}, \{8, 9, 12\}.
 \end{aligned}$$

These are $26 = 13 \cdot 12/6$ triples covering each of the 78 pairs of $\mathbb{Z}_{13}$ exactly once.

For the doubling, let $W = \{w_0, \dots, w_{12}, w_\infty\}$ and take the standard 1-factorization of the complete graph on W indexed by $\mathbb{Z}_{13}$,

$$F_i = \{\{w_\infty, w_i\}\} \cup \{\{w_{i+j}, w_{i-j}\} : j = 1, \dots, 6\} \quad (i \in \mathbb{Z}_{13}),$$

subscripts modulo 13. Each F_i is a perfect matching of the 14 points of W , and the thirteen F_i partition all 91 edges, because $\{w_a, w_b\}$ with $a \neq b$ lies in F_i for the unique $i = (a + b)/2$ (2 being invertible modulo 13) and $\{w_\infty, w_a\}$ lies in F_a alone. Let D be the STS(27) on $\mathbb{Z}_{13} \cup W$ whose block set is

$$(2) \cup \{\{i\} \cup e : i \in \mathbb{Z}_{13}, e \in F_i\},$$

that is, the 26 blocks of (2) together with the 91 blocks $\{i, w_\infty, w_i\}$ and $\{i, w_{i+j}, w_{i-j}\}$ for $i \in \mathbb{Z}_{13}$ and $j = 1, \dots, 6$; in all $26 + 91 = 117 = 27 \cdot 26/6$ blocks. Finally T is the STS(39) on $\mathbb{Z}_{13} \times \mathbb{Z}_3$ given by the tripling rule of §1 applied to (2), with $13 + 9 \cdot 26 = 247 = 39 \cdot 38/6$ blocks.

3. PROOF OF THEOREM 1

(a) H is non-degenerate. A proper subsystem S of an STS(v) has order w with $v \geq 2w + 1$: fix a point $x \notin S$ and send $p \in S$ to the third point $f(p)$ of the block $\{x, p, f(p)\}$. If $f(p) = f(q)$ for $p \neq q$ then the block through $f(p)$ and x would meet S twice, forcing $x \in S$; and $f(p) \in S$ would likewise force $x \in S$. So f is an injection of S into $V \setminus (S \cup \{x\})$, whence $w \leq v - w - 1$. For $v = 13$ this gives $w \leq 6$, so the only admissible orders of a proper subsystem are $w = 1$ and $w = 3$: H contains no subsystem of order 7 or 9. The subsystems of order 3 are exactly the blocks, and each block is a maximal proper subsystem. A block, as an STS(3), is generated by any two of its points and by no single point, so $b_L = 2$ there. On the other hand no pair of points generates H , since the closure of a pair is the block through it, while H is generated by three points: the closure of $\{0, 1, 2\}$ already contains $\{0, 1, 2, 4, 5, 7\}$, the blocks through the three pairs being $\{0, 1, 4\}$,

$\{1, 2, 5\}$ and $\{0, 2, 7\}$, and a subsystem of order at least 6 must have order 13, the admissible orders being only 1, 3 and 13. Hence $b_L(H) = 3$ and every maximal proper subsystem S has $b_L(S) = 2 = b_L(H) - 1$: H is non-degenerate.

(b) H is neither projective nor affine. $2^{n+1} - 1$ takes the values 7 and 15 on either side of 13, and 3^m takes the values 9 and 27. Both are statements about the *order*, so they hold for every STS(13) and require no isomorphism test.

(c) The burning numbers. Define, with $\binom{\cdot}{2}$ the binomial coefficient,

$$(4) \quad h(1) = 1, \quad h(r) = \binom{h(r-1)}{2} - 2h(r-1) + 3r - 2,$$

so that $h(1), \dots, h(6) = 1, 2, 4, 8, 25, 266$. Write ρ_v for the least r with $v \leq h(r)$, and $U(v) = \lceil \log_2 v \rceil + 1$. Two theorems of [1] are unconditional and hold for every Steiner triple system:

$$(5) \quad \rho_v \leq b(K) \leq U(v) \quad \text{for every STS}(v) \text{ } K.$$

The lower bound is the theorem of [1] whose engine is the statement that at most $h(r)$ points can be alight at the end of round r ; the upper bound is the bound $b \leq \lceil \log_2 v \rceil + 1$ proved there for all Steiner triple systems. Now

$$\begin{aligned} h(4) = 8 < 13 \leq 25 = h(5) &\Rightarrow \rho_{13} = 5, \quad U(13) = 4 + 1 = 5; \\ h(5) = 25 < 27 \leq 266 = h(6) &\Rightarrow \rho_{27} = 6, \quad U(27) = 5 + 1 = 6; \\ h(5) = 25 < 39 \leq 266 = h(6) &\Rightarrow \rho_{39} = 6, \quad U(39) = 6 + 1 = 7. \end{aligned}$$

In the first two lines the two bounds of (5) coincide, so $b(K) = 5$ for *every* STS(13) and $b(K) = 6$ for *every* STS(27). In particular $b(H) = 5$ and $b(D) = 6$ whatever 1-factorization is used in (3), so the “for some image” and “for every image” readings of “increases on doubling” agree here. Finally $b(T) \geq \rho_{39} = 6 > 5 = b(H)$. This settles (c), and with it Theorem 1, by counting alone.

4. WHAT IS NOT SETTLED

The two equalities used in (c) hold for every system of order 13 and every system of order 27; they are facts about those orders and not about H . So the witness contradicts the question as printed without bearing on the structural phenomenon that [1] exhibits at the projective and affine systems, and nothing here shows that a projective/affine characterisation fails at orders where the two bounds of (5) do not coincide. A referee may reasonably read Theorem 1 as a small-order artefact of those bounds.

The increase from H to D is by exactly one, both endpoints being pinned. Whether the increase from H to T is by exactly one is not determined here: (5) confines $b(T)$ to $\{6, 7\}$. The reading on which the question asks for an increase *by one* on doubling *and* on tripling is therefore not settled by this witness.

5. VERIFICATION

The program accompanying this note reads the 26 blocks of (2) as they are printed above and re-derives the quantities asserted here: that (2) is an STS(13) and coincides with the cyclic development of $\{0, 1, 4\}$ and $\{0, 2, 7\}$; that the F_i are a 1-factorization and that D and T are Steiner triple systems of orders 27 and 39 restricting to H as claimed; that H has exactly 41 subsystems, that its 26 maximal proper subsystems are its blocks, and that $b_L(H) = 3$ with $b_L = 2$ on each of them, so that H is non-degenerate; the recursion (4) and the values $\rho_{13}, \rho_{27}, \rho_{39}$ and $U(13), U(27), U(39)$, with $\lceil \log_2 v \rceil$ evaluated exactly by bit length rather than by a floating-point logarithm. It also recomputes $b(H) = 5$, $b(D) = 6$ and $b(T) > 5$ from the definition (1) alone, by enumerating every burnt set reachable at the end of each round, so that (c) does not rest on (5); the sources 0, 1, 2, 3, 6 burn H in five rounds. As controls it reproduces $b(\text{STS}(3)) = 3$, $b(\text{PG}(2, 2)) = 4$, $b(\text{AG}(2, 3)) = 5$, $b(\text{PG}(3, 2)) = 5$ and $b(\text{AG}(3, 3)) = 6$, all values published in [1], and checks that the tripling rule carries $\text{AG}(2, 3)$ to $\text{AG}(3, 3)$. It is Python 3.9 or later, standard library only, exact integer arithmetic throughout, and it runs in a few seconds. Its transcript accompanies it.

Its scope has limits, which it also prints. It does not fetch or parse the e-print of [1]: the quoted question is transcribed from the source, and only the values of h are recomputed. It does not consult [3]; the doubling and tripling rules are implemented from the statements reproduced in §1. No isomorphism census is performed. The exact value of $b(T)$, which is 6 or 7, is not determined.

REFERENCES

- [1] A. C. Burgess, P. H. Danziger, C. W. Jones, T. G. Marbach, and D. A. Pike, *Burning Steiner triple systems*, arXiv:2608.25627v1, 2026. arXiv:2608.25627. The question refuted here is unnumbered in that e-print; it is quoted in full in §1.
- [2] A. C. Burgess, C. W. Jones, and D. A. Pike, *Extending graph burning to hypergraphs*, Ars Math. Contemp. **26** (2026), Paper P3.07. arXiv:2403.01001.
- [3] C. J. Colbourn and J. H. Dinitz (eds.), *Handbook of Combinatorial Designs*, 2nd ed., Chapman & Hall/CRC, Boca Raton, 2007. Constructions 2.15 (double-and-add-one) and 2.17 (tripling). Not consulted directly; the two constructions are used here in the standard form reproduced in §1.